

Advanced Time Series Analysis and Forecasting

Chapter 4: Cointegration revisited: VECM, ARDL and panel data



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Acronyms used in this material (1/2)

- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **ARDL**: AutoRegressive Distributed Lag (model)
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **CADF**: Cross-sectionally augmented Dickey–Fuller (regression)
- **CCE**: Common Correlated Effects (estimator)
- **CCEMG**: Common Correlated Effects Mean Group
- **CD**: Cross-section Dependence (test of Pesaran)
- **CIPS**: Cross-sectionally augmented IPS (test)
- **CS-ARDL**: Cross-Sectionally augmented ARDL
- **DOLS**: Dynamic Ordinary Least Squares
- **ECM**: Error Correction Model
- **EU**: European Union
- **EU-27**: the 27 member states of the European Union
- **FE**: Fixed Effects
- **FEVD**: Forecast Error Variance Decomposition
- **FMOLS**: Fully Modified Ordinary Least Squares
- **FRED**: Federal Reserve Economic Data
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity

Acronyms used in this material (2/2)

- **GDP**: Gross Domestic Product
- **GMM**: Generalized Method of Moments
- **HICP**: Harmonised Index of Consumer Prices
- **IMF**: International Monetary Fund
- **IPS**: Im–Pesaran–Shin (panel unit-root test)
- **LLC**: Levin–Lin–Chu (panel unit-root test)
- **LLM**: Large Language Model
- **LR**: Likelihood Ratio (test)
- **MG**: Mean Group (estimator)
- **ML**: Machine Learning
- **NARDL**: Nonlinear AutoRegressive Distributed Lag (model)
- **NPISH**: Non-Profit Institutions Serving Households
- **OECD**: Organisation for Economic Co-operation and Development
- **OLS**: Ordinary Least Squares
- **PMG**: Pooled Mean Group (estimator)
- **PPP**: Purchasing Power Parity
- **PSS**: Pesaran–Shin–Smith
- **ROBOR**: Romanian Interbank Offer Rate
- **SE**: Standard Error
- **TSA**: Time Series Analysis
- **US**: United States
- **VAR**: Vector AutoRegression
- **VECM**: Vector Error Correction Model

Today's question and route

- **Question**: when do non-stationary series share a long-run equilibrium, which restrictions on it can we test, and how far can we trust the answer in samples of 100–300 observations or across 27 countries?
  - cointegration is a statement about the likelihood of a reduced-rank VAR; inference on it is non-standard and fragile
- **Route** of the chapter
  - the cointegrated VAR as a likelihood problem; the five deterministic cases; small-sample corrections and the bootstrap
  - identification and tests on β and α; interest-rate pass-through in Romania; I(2) in brief; common trends
  - ARDL and the bounds test; panel unit roots, panel cointegration and heterogeneous panel estimators
- We build on TSA, Chapter 7 (Engle–Granger, ECM, the VECM, Johansen tests, pairs trading) and on Chapter 3 (structural identification); Seminar 4 comes before this lecture

Learning outcomes

- Derive the Johansen estimator as a reduced-rank regression, state the asymptotic distribution of the trace statistic and choose the deterministic case
- Correct the rank test in small samples (Reinsel–Ahn, Bartlett, wild bootstrap) and report the sensitivity of the rank to the lag length
- Identify cointegrating vectors, test economic restrictions on β and weak exogeneity through α, and build a common-trends structural VECM
- Run the ARDL bounds test with correct (and small-sample) critical values, and decide between ARDL and VECM
- Test for cross-section dependence, panel unit roots and panel cointegration, and estimate long-run panel relations by MG, PMG and CCE

Reading, data and tools

- Backbone: Johansen (1995a); Juselius (2006); Kilian and Lütkepohl (2017), Ch. 3 and 10; Pesaran (2015b), Ch. 22–31; Lütkepohl (2005), Ch. 6–9
  - original papers: Johansen (1988), Johansen (1991), Pesaran, Shin and Smith (2001), Pesaran, Shin and Smith (1999), Pesaran (2006), Pesaran (2007); survey: Breitung and Pesaran (2008)
- Python Quantlets of this chapter: Quantlets/Ch_04
  - Johansen in five cases, restricted ML, bootstrap rank tests, ARDL bounds, LLC/IPS/CIPS, Pedroni, Westerlund, MG, PMG and CCE written out in `numpy`
- Lecture notebook: open in Google Colab

Data used in this chapter

Series	Source	Use
Romanian lending and deposit rates (lei); ROBOR 3M	IMF International Financial Statistics (BNR data); Eurostat irt_st_m	VECM, ARDL
US consumption of nondurables and services, fixed investment, GDP, deflator, population	FRED (PCND, PCESV, FPI, GDP, GDPDEF, B230RC0Q173SBEA)	common trends
Romanian HICP, monthly	Eurostat prc_hicp_minr	I(2)
EU-27 household consumption, disposable income, consumption deflator, population	Eurostat nama_10_gdp, nasa_10_nf_tr, nama_10_pe	panel

- All sources are public and need no account or key; monthly rates to March 2026, US data to 2025, EU annual data to 2024
- HICP: harmonised index of consumer prices; NPISH: non-profit institutions serving households (included with households)

Origins of likelihood-based cointegration analysis



Robert Engle, 2017

Photo: Econterms (2017); CC BY-SA 4.0; Wikimedia Commons

- 1987: Engle and Granger (1987) define cointegration and prove the representation theorem; estimation is a two-step regression
- 1988–1991: Johansen (1988) and Johansen (1991) turn it into maximum likelihood for a VAR with reduced rank: several relations, tests on their coefficients
- 1999–2001: Pesaran, Shin and Smith add ARDL bounds tests Pesaran, Shin and Smith (2001) and pooled panels Pesaran, Shin and Smith (1999)
- 2003: Engle and Granger share the Nobel Prize; Granger "for methods of analyzing economic time series with common trends (cointegration)"
- Since then: bootstrap rank tests, cross-section dependence in panels, common factors

Prior knowledge from TSA and new material

- Known (TSA, Chapter 7): $y_t \sim \mathrm{I}(1)$, $\beta' y_t \sim \mathrm{I}(0)$; Engle–Granger two-step; the ECM; $\Pi = \alpha\beta'$; trace and max-eigenvalue tests with tabulated critical values
- New here
  - why the eigenvalue problem is the maximum likelihood estimator, and what its limit distribution looks like
  - which deterministic case to use, and what goes wrong in samples of 100–300 months
  - restrictions on β (economic hypotheses) and on α (weak exogeneity); structural shocks with permanent effects
  - single equations (ARDL) and panels of countries when the system approach is too demanding

The VECM and the hypothesis $H(r)$ (1/2)

- The VECM: this month's change of each variable responds to last month's levels, to past changes and to deterministic terms

$$\Delta y_t = \Pi y_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta y_{t-i} + \Phi D_t + \varepsilon_t$$

- Notation
  - $y_t \in \mathbb{R}^n$: the vector of the n I(1) variables; $\Delta y_t = y_t - y_{t-1}$: their changes
  - Π $(n \times n)$: the long-run (levels) matrix; Γ_i $(n \times n)$: the short-run matrices; p: the lag order of the VAR in levels
  - D_t: deterministic terms (constant, trend, dummies) with coefficients Φ
  - $\varepsilon_t \sim$ i.i.d. $N(0, \Omega)$: the innovations, with covariance matrix Ω
- Gaussian likelihood conditional on the first p observations: the estimator is ML, the tests are likelihood ratios Johansen (1988), Johansen (1991)

The VECM and the hypothesis $H(r)$ (2/2)

- $H(r)$: Π has rank at most r, so it factors as $\Pi = \alpha\beta'$ with α, β of dimension $n \times r$
  - r: the number of cointegrating relations, $0 \leq r \leq n$
  - β: its columns are the r cointegrating vectors; $\beta' y_t$ is stationary (the long run)
  - α: the adjustment (loading) coefficients; α_{ij} is the response of $\Delta y_{i,t}$ to the deviation $\beta_j' y_{t-1}$
- The models are nested: $H(0) \subset H(1) \subset \cdots \subset H(n)$
  - $H(n)$: unrestricted VAR in levels; $H(0)$: VAR in differences
- Only Π is identified, not α and β separately: $\alpha\beta' = (\alpha\xi)(\beta\xi^{-1\prime})'$
  - ξ: any invertible $r \times r$ matrix; $\xi^{-1\prime}$: the transpose of its inverse

Step 1: concentrating out the short run

- Regress Δy_t and y_{t-1} on $Z_{2t} = (\Delta y'_{t-1}, \ldots, \Delta y'_{t-p+1}, D'_t)'$ and keep the residuals (Frisch–Waugh)
  - Z_{2t}: all short-run regressors (lagged changes and deterministic terms)
  - R_{0t}, R_{1t}: the residuals of Δy_t and of y_{t-1}, i.e. both purged of the short run
- The concentrated model is a regression with a reduced-rank coefficient: $R_{0t} = \alpha\beta' R_{1t} + \hat{\varepsilon}_t$
- Moment matrices $S_{ij} = T^{-1} \sum_t R_{it} R'_{jt}$, $i, j \in \{0, 1\}$
  - T: the number of observations used; S_{00}, S_{11}: the covariance matrices of R_{0t}, R_{1t}; $S_{01} = S'_{10}$: their cross-covariance
- For fixed β, OLS of R_{0t} on $\beta' R_{1t}$ gives α and Ω in closed form:
  - $\hat{\alpha}(\beta) = S_{01}\beta(\beta' S_{11}\beta)^{-1}$, $\hat{\Omega}(\beta) = S_{00} - S_{01}\beta(\beta' S_{11}\beta)^{-1}\beta' S_{10}$
  - $L_{\max}(\beta)$: the likelihood maximised over all parameters except β; $L_{\max}^{-2/T}(\beta) \propto |\hat{\Omega}(\beta)|$ ($|\cdot|$: determinant)
  - what remains is a problem in β alone: minimise $|\hat{\Omega}(\beta)|$

Step 2: the eigenvalue problem

- Determinant identity: $|\hat{\Omega}(\beta)| = |S_{00}| \dfrac{|\beta'(S_{11} - S_{10}S_{00}^{-1}S_{01})\beta|}{|\beta' S_{11}\beta|}$
- Minimising this ratio of quadratic forms is the generalised eigenvalue problem $|\lambda S_{11} - S_{10}S_{00}^{-1}S_{01}| = 0$
  - its n roots, ordered, are the eigenvalues $1 > \hat{\lambda}_1 > \cdots > \hat{\lambda}_n > 0$; $\hat{v}_i$: the eigenvector of $\hat{\lambda}_i$
- $\hat{\beta}$ = the eigenvectors $\hat{v}_1, \ldots, \hat{v}_r$ of the r largest eigenvalues; then $L_{\max}^{-2/T} = |S_{00}| \prod_{i=1}^{r}(1 - \hat{\lambda}_i)$
  - normalisation $\hat{\beta}' S_{11} \hat{\beta} = I_r$ (I_r: the $r \times r$ identity matrix)
- $\hat{\lambda}_i$ are the squared canonical correlations between R_{0t} and R_{1t}: large $\hat{\lambda}_i$ = a linear combination of levels that predicts the changes
- One eigen-decomposition gives the ML estimates for every r at once (proof: Appendix)

Likelihood ratio tests of the rank (1/2)

- **Trace**: $H(r)$ against $H(n)$: $LR_{tr}(r) = -T \sum_{i=r+1}^{n} \ln(1 - \hat{\lambda}_i)$
  - sums the $n - r$ smallest eigenvalues, those that $H(r)$ sets to zero
  - $\hat{\lambda}_i \approx 0$ gives $-\ln(1 - \hat{\lambda}_i) \approx 0$: a small statistic supports $H(r)$; large values reject it
- **Maximum eigenvalue**: $H(r)$ against $H(r+1)$: $LR_{\max}(r) = -T \ln(1 - \hat{\lambda}_{r+1})$
  - uses only the next eigenvalue, $\hat{\lambda}_{r+1}$: is there one more relation?
- Rank determination: test $r = 0, 1, \ldots$ in turn and stop at the first non-rejection; this sequence is consistent for the true rank at a fixed level Johansen (1995a)

Likelihood ratio tests of the rank (2/2)

- Limit under $H(r)$:

$$LR_{tr} \Rightarrow \mathrm{tr}\Big\{ \int_0^1 (dB)F' \Big(\int_0^1 FF'du \Big)^{-1} \int_0^1 F(dB)' \Big\}$$

- Notation
  - $\Rightarrow$: convergence in distribution as $T \to \infty$; tr: the trace (sum of the diagonal elements)
  - $B(u)$, $u \in [0,1]$: an $(n-r)$-dimensional standard Brownian motion, the limit of the rescaled common trends
  - $F = B$ corrected for the deterministic terms of the case
- The limit depends only on $n-r$ and the case, not on $\alpha, \beta, \Gamma_i, \Omega$
  - a multivariate Dickey–Fuller distribution: quantiles by simulation Osterwald-Lenum (1992), MacKinnon, Haug and Michelis (1999)
  - non-standard: neither χ^2 nor Normal

The Granger representation theorem (1/3)



Clive Granger, 2008

Photo: Olaf Storbeck (2008); CC BY-SA 2.0; Wikimedia Commons

- The theorem writes a cointegrated VAR as random-walk trends plus stationary deviations
- VAR in levels: $y_t = \sum_{i=\mathbf{1}}^{p} A_i y_{t-i} + \Phi D_t + \varepsilon_t$
  - A_i $(n \times n)$: its lag matrices; $A(z) = I - \sum_{i=\mathbf{1}}^{p} A_i z^i$: its characteristic polynomial
  - $\Pi = -A(1)$, $\Gamma_i = -\sum_{j>i} A_j$: the VECM matrices; $\Gamma = I - \sum_i \Gamma_i$
- Conditions
  - the roots of $|A(z)| = 0$ satisfy $|z| > 1$ or $z = 1$ (no explosive or seasonal roots)
  - $\operatorname{rank} \Pi = r$, so $\Pi = \alpha\beta'$
  - $\alpha_\perp' \Gamma \beta_\perp$ has full rank $n - r$
  - $\alpha_\perp$, $\beta_\perp$ $(n \times (n-r))$: orthogonal complements, $\alpha'\alpha_\perp = 0$, $\beta'\beta_\perp = 0$

The Granger representation theorem (2/3)

- Under these conditions y_t is I(1), $\beta' y_t$ is I(0), and

$$y_t = C \sum_{i=1}^{t} (\varepsilon_i + \Phi D_i) + C^*(L)(\varepsilon_t + \Phi D_t) + A_0$$

- Notation
  - $C = \beta_\perp (\alpha'_\perp \Gamma \beta_\perp)^{-1} \alpha'_\perp$: the long-run impact matrix, of rank $n - r$ (derivation: Appendix)
  - $C^*(L) = \sum_{j \geq 0} C_j^* L^j$: a polynomial in the lag operator L ($L y_t = y_{t-1}$) with summable coefficients, the stationary part
  - A_0: a constant that depends on the initial values, with $\beta' A_0 = 0$
- First term: random walks plus deterministic trends; second term: stationary deviations from them

The Granger representation theorem (3/3)

- Interpretation
  - $\beta' C = 0$: the relations do not contain the stochastic trends
  - $C\alpha = 0$: an equilibrium error has no permanent effect
  - common trends: the $n - r$ random walks $\alpha'_\perp \sum_i \varepsilon_i$
- If $\alpha'_\perp \Gamma \beta_\perp$ is singular, the theorem fails
  - C does not exist and some trends are I(2) (see the I(2) section)

Recap: The cointegrated VAR

- Concentrate out the short run, then maximise over β: a generalised eigenvalue problem
- The eigenvalues are squared canonical correlations; the trace and max-eigenvalue statistics are likelihood ratios
- Their limits are Brownian functionals that depend on $n - r$ and on the deterministic terms only
- Granger representation: $C = \beta_\perp(\alpha_\perp'\Gamma\beta_\perp)^{-1}\alpha_\perp'$ links α, β to the common trends

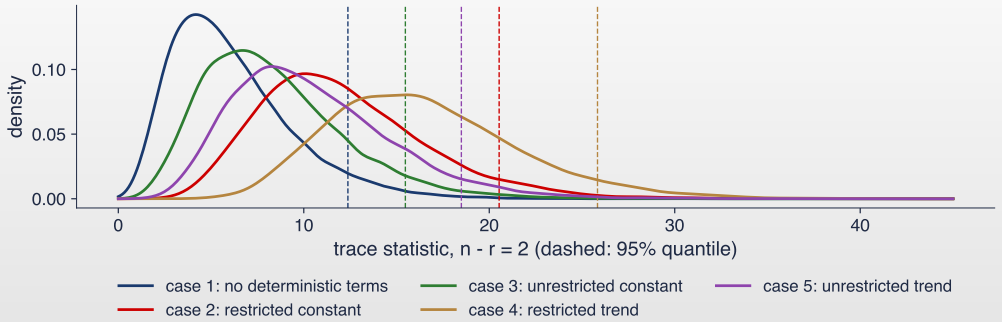
Five ways to place a constant and a trend

- $\mu_t = \mu_0 + \mu_1 t$ with $\mu_j = \alpha\rho_j + \alpha_\perp\gamma_j$: the α part sits in the relations, the $\alpha_\perp$ part drives the levels Johansen (1995a)
  - μ_t: constant μ_0 plus trend $\mu_1 t$ in the VECM; ρ_j $(r \times 1)$, γ_j $((n-r) \times 1)$: their coordinates in the two directions

Case	Restriction	Levels y_t	Relations $\beta' y_t$
1 (H_2)	$\mu_0 = \mu_1 = 0$	no drift	zero mean
2 (H_1^*)	$\mu_0 = \alpha\rho_0$, $\mu_1 = 0$	no drift	constant
3 (H_1)	$\mu_1 = 0$	linear trend	constant
4 (H^*)	$\mu_1 = \alpha\rho_1$	linear trend	linear trend
5 (H)	none	quadratic trend	linear trend

- "Restricted" terms enter Z_{1t} (the regressors multiplied by Π) with y_{t-1}, as part of β; "unrestricted" terms enter Z_{2t} and are concentrated out
- Interest rates: case 2; trending macro aggregates: case 3 or 4; case 5 is rarely plausible

The deterministic case and the null distribution



- Trace statistic under $H(r)$ with $n - r = 2$, simulated from random walks (10 000 replications, $T = 400$; drift in cases 3 and 5)
- 95% quantiles for every case and $n - r = 1, \ldots, 4$: the table two slides later

ATS_ch4_johansen_asymptotics

Interpreting the five distributions

- Each restricted deterministic term adds a regressor to the reduced-rank problem and shifts the distribution to the right
- A trend in the levels (cases 3 and 5) makes one direction of the common trend deterministic: for $n - r = 1$ the limit is $\chi^2(1)$
- Using case-3 critical values in a case-2 model over-rejects: the 95% quantile for $n - r = 1$ is 9.25 in case 2 and 3.89 in case 3

Simulated 95% quantiles of the trace statistic

$n - r$	case 1	case 2	case 3	case 4	case 5
1	4.21	9.25	3.89	12.58	3.84
2	12.38	20.53	15.48	25.84	18.50
3	24.44	35.24	29.78	43.14	35.36
4	40.29	53.68	47.98	63.85	55.54

- Simulated with 10 000 replications of a VAR(1) test regression on $T = 400$ observations; the response-surface critical values of MacKinnon, Haug and Michelis (1999) differ by at most a few tenths
- Choosing the case: plot the data first; if unsure between 2 and 3 (or 3 and 4), test the joint hypothesis on rank and case in the order of the Pantula principle Pantula (1989), Johansen (1992)
- Breaks in the deterministic terms shift the distribution again: use the critical values of Johansen, Mosconi and Nielsen (2000) for broken trends

Recap: Deterministic terms

- Split each deterministic term into an α part (in the relations) and an $\alpha_\perp$ part (in the levels)
- The five cases have five different null distributions; mixing them up changes the rank
- Plot the data, choose the case on economic grounds and use the Pantula principle when in doubt

Over-rejection of the asymptotic test (1/2)

- The VECM spends many parameters on the short run
  - $n^2(p-1)$ elements of $\Gamma_1, \ldots, \Gamma_{p-1}$, plus the deterministic terms
  - with $T = 100$, $n = 3$ and $p = 4$: 27 short-run parameters, a large share of the information
- The distortion grows with persistence
  - persistent short-run dynamics: roots of the stationary part close to the unit circle
  - the asymptotic test then finds too many cointegrating relations
- Conditional heteroskedasticity Cavaliere, Rahbek and Taylor (2010)
  - leaves the limit distribution unchanged
  - but worsens the finite-sample size: use the wild bootstrap

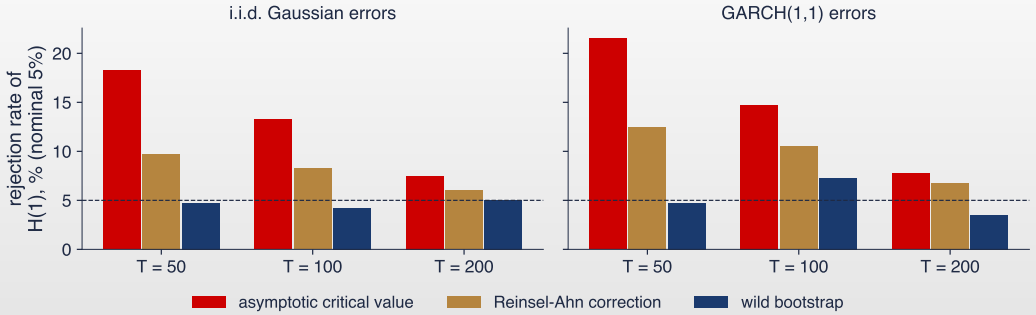
Over-rejection of the asymptotic test (2/2)

- **Reinsel–Ahn** Reinsel and Ahn (1992)
  - multiply the statistic by $(T - np)/T$: a degrees-of-freedom correction
  - simple, often too conservative
- **Bartlett correction** Johansen (2002)
  - $\mathbb{E}LR_{tr} \approx f\,(1 + a(\theta)/T)$; the corrected statistic is $LR_{tr}/(1 + a(\theta)/T)$
  - f: the mean of the limit distribution; θ: the VECM parameters
  - $a(\theta)$: an analytic function of θ, evaluated at the estimates
- **Bootstrap** Swensen (2006), Cavaliere, Rahbek and Taylor (2012)
  - simulate the null distribution from the model estimated under $H(r)$

The bootstrap rank test, step by step

- 1. Estimate the VECM under $H(r)$: $\hat{\alpha}^{(r)}, \hat{\beta}^{(r)}, \hat{\Gamma}_i^{(r)}, \hat{\Phi}^{(r)}$ and residuals $\hat{\varepsilon}_t^{(r)}$
- 2. Generate $\Delta y_t^* = \hat{\alpha}^{(r)}\hat{\beta}^{(r)\prime} y_{t-1}^* + \sum_i \hat{\Gamma}_i^{(r)} \Delta y_{t-i}^* + \hat{\Phi}^{(r)} D_t + \varepsilon_t^*$ from the observed initial values
- 3. Innovations: $\varepsilon_t^* = \hat{\varepsilon}_t^{(r)} w_t$, $w_t \sim$ i.i.d. $N(0,1)$ (wild) or resampled residuals (i.i.d. bootstrap)
  - the star marks bootstrap quantities; w_t: a random sign-and-scale weight; the wild version keeps the volatility pattern of each date
- 4. Compute $LR_{tr}^*(r)$ on each sample; p-value = share of $LR_{tr}^* \geq LR_{tr}(r)$; test $r = 0, 1, \ldots$ and stop at the first non-rejection
- Estimating under $H(r)$ (not under $H(n)$) is essential: the bootstrap data must have exactly $n - r$ unit roots Cavaliere, Rahbek and Taylor (2012)

Size of the rank test in small samples (simulation)



- Three-variable VECM, true rank 1, $\Gamma_1 = 0.8I$, case 2, VAR(2); test of $H(1)$ at 5%; 400 replications, 199 bootstrap samples each
- $T = 50$, i.i.d. errors: asymptotic 18.2%, Reinsel–Ahn 9.8%, wild bootstrap 4.8%

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Interpreting the size simulation

- With persistent short-run dynamics the asymptotic test finds a spurious second relation in 18.2% of samples of 50 and 13.2% of samples of 100
- Reinsel–Ahn halves the distortion; the wild bootstrap is close to 5% at every T, also under GARCH errors (4.8% at $T = 50$, asymptotic 21.5%)
- At $T = 200$ all three are acceptable: the problem is the ratio of parameters to observations, not the method
- Practice: report asymptotic and bootstrap p-values side by side, and the rank for neighbouring lag lengths

Recap: Small samples

- The asymptotic trace test over-rejects when parameters are many and the short run is persistent
- Reinsel–Ahn and the Bartlett correction rescale the statistic; the bootstrap rebuilds its distribution
- Bootstrap from the model estimated under $H(r)$; use wild weights when volatility changes over time

Normalisation and identification of the cointegrating vectors (1/2)

- The eigenvectors give one basis of the cointegration space $\mathrm{sp}(\beta)$; any $\beta\xi$ spans the same space and fits equally well
  - $\mathrm{sp}(\beta)$: the space spanned by the columns of β; ξ: an invertible $r \times r$ matrix (a rotation)
- Dividing each vector by one coefficient fixes the scale (r restrictions), but not the rotation
  - $r(r-1)$ further restrictions are needed, $r-1$ per vector
- Identification rests on economic theory: each restriction is a statement such as "the lending rate does not depend on the deposit rate in the long run"

Normalisation and identification of the cointegrating vectors (2/2)

- Linear restrictions vector by vector: $\beta = (H_1\varphi_1, \ldots, H_r\varphi_r)$
  - n_1: the length of each vector (n plus the restricted deterministic terms)
  - H_i ($n_1 \times s_i$, known): the design matrix of vector i; φ_i ($s_i \times 1$): its s_i free coefficients
  - $R_i = H_{i\perp}$: the restrictions written as $R_i'\beta_i = 0$
- Rank condition Johansen (1995b): for every i and every set of k other vectors $i_1, \ldots, i_k$
  - $\operatorname{rank}(R_i'(H_{i_1}, \ldots, H_{i_k})) \geq k$, $k = 1, \ldots, r-1$
  - meaning: no combination of other vectors satisfies the restrictions of vector i
  - the cointegration analogue of the rank condition in simultaneous equations; the order condition alone is not enough

Testing restrictions on β (1/2)

- **The same restriction on all vectors**, $\beta = H\varphi$: solve $|\lambda H' S_{11} H - H' S_{10} S_{00}^{-1} S_{01} H| = 0$
  - H ($n_1 \times s$, known): the restriction; φ ($s \times r$): the free coefficients; e.g. excluding one variable from all relations
  - $\tilde{\lambda}_i$: the eigenvalues of the restricted problem; $\hat{\lambda}_i$: the unrestricted ones
- Test statistic: $LR = T \sum_{i=1}^{r} \ln\{(1-\tilde{\lambda}_i)/(1-\hat{\lambda}_i)\} \to \chi^2(r(n_1 - s))$
  - $LR \geq 0$; it is large when the restriction lowers the canonical correlations, i.e. when the data reject it
  - degrees of freedom $r(n_1 - s)$: $n_1 - s$ restrictions on each of the r vectors
- Given the rank, $\hat{\beta}$ is super-consistent and mixed Gaussian: LR tests on β are χ^2, unlike the rank test Johansen (1991)

Testing restrictions on β (2/2)

- **Different restrictions per vector**, $\beta = (H_1\varphi_1, \ldots, H_r\varphi_r)$: no closed form
  - maximise the concentrated log-likelihood $-\frac{T}{2}\ln|\hat{\Omega}(\beta)|$ by the switching algorithm (one vector at a time) or numerically
  - over-identifying restrictions: $LR \to \chi^2(\sum_i (n_1 - r + 1 - s_i))$; each vector has $n_1 - r + 1 - s_i$ restrictions beyond the $r - 1$ needed to identify it
- **Fully known** $\beta = \beta_0$ (e.g. the great ratios):

$$LR = T\Big\{\ln|\hat{\Omega}(\beta_0)| - \ln|S_{00}| - \sum_{i\le r}\ln(1-\hat{\lambda}_i)\Big\} \to \chi^2(r(n_1 - r))$$

  - compares the fit with β_0 imposed ($\hat{\Omega}(\beta_0)$) to the unrestricted fit; β_0: the hypothesised matrix

Restrictions on α and weak exogeneity

- $\alpha = A\psi$: only m combinations of the equations adjust to the relations
  - A ($n \times m$, known): the restriction; ψ ($m \times r$): the free adjustment coefficients; $A_\perp$: its orthogonal complement
  - condition the system on $A'_\perp R_{0t}$ and solve the eigenvalue problem of the remaining m equations (eigenvalues $\tilde{\lambda}_i$)
  - $LR = T \sum_{i \leq r} \ln\{(1 - \tilde{\lambda}_i)/(1 - \hat{\lambda}_i)\} \to \chi^2(r(n-m))$
- **Weak exogeneity** of x_t for β: the row of α for x_t is zero (a zero row of α, r restrictions)
  - x_t does not adjust to past disequilibria; it is a common trend (a column of $\alpha_\perp$)
  - the conditional model of Δy_t given Δx_t is then fully efficient for β: the basis of single-equation ECM and ARDL
- Long-run structural hypotheses are joint restrictions on β and α
  - PPP: $(1, -1, -1)$ on (price, foreign price, exchange rate); Fisher: $(1, -1)$ on (interest rate, inflation); complete pass-through: $(1, -1)$ on (retail rate, policy or market rate)

Recap: Identification and tests

- Identify β with $r-1$ economically motivated restrictions per vector; check the rank condition
- Given the rank, LR tests on β and α are χ^2 with known degrees of freedom
- A zero row of α = weak exogeneity = permission to condition on that variable

Speed and completeness of the pass-through from ROBOR

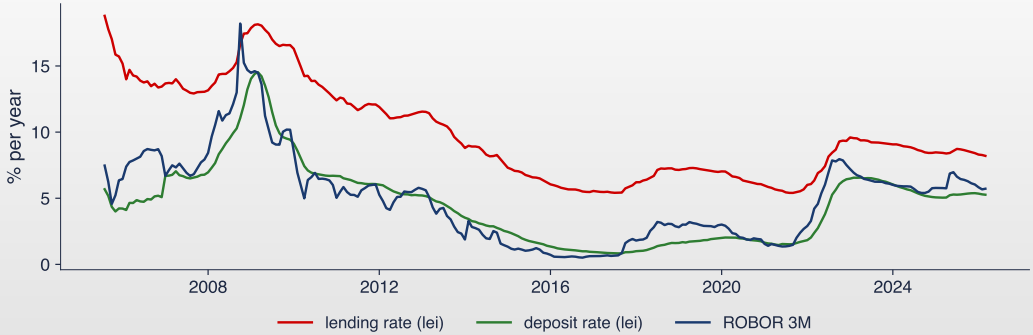


The BNR palace, Bucharest

Photo: Ștefan Jurcă (2015); CC BY 2.0; Wikimedia Commons

- Monetary policy reaches households and firms only through bank rates; ROBOR 3M is the reference for most variable-rate loans in lei
- Euro area: long-run pass-through to lending rates often complete, to deposit rates incomplete de Bondt (2005); CEE: faster and fuller pass-through after the early transition Égert, Crespo-Cuaresma and Reininger (2007)
- Hypotheses: H_β: lending and deposit rates move one for one with ROBOR in the long run; H_α: ROBOR does not adjust to bank rates
- Data: August 2005 – March 2026, $T = 248$ months (inflation targeting)

Lending rate, deposit rate and ROBOR 3M



- Monthly, % per year; lending and deposit rates in lei (IMF IFS, reported by the BNR); ROBOR 3M (Eurostat)
- ROBOR ranges from 0.51% to 18.2%; the lending–ROBOR spread falls from 11.3 pp to 2.5 pp

ATS_ch4_passthrough_vecm

Interpreting the three rates

- All three rates wander far from any mean over twenty years: treat them as I(1) within the sample, even if interest rates are bounded in theory
- They move together after 2008, in 2014–2015 and in 2022: the visual case for cointegration
- The falling spread in 2005–2010 is the end of the transition (competition, falling risk premia), not a policy effect
- No trend in levels is plausible for rates: case 2, constant restricted to the relations

Rank of the pass-through system

$H(r)$	LR_{tr}	RA	cv 5%	wild p	LR_{tr}, p=4	RA, p=4	wild p, p=4
$r = 0$	121.1	118.1	35.2	0.007	40.0	38.0	0.647
$r \leq 1$	52.2	50.9	20.5	0.022	14.9	14.1	0.812
$r \leq 2$	4.3	4.2	9.3	0.767	3.8	3.6	0.820

- Case 2; RA: Reinsel–Ahn; cv: simulated 95% quantile; wild p: 399 wild-bootstrap samples under $H(r)$; lag order by BIC $= 2$, HQ $= 5$, AIC $= 8$
- With $p = 2$ all three methods give $r = 2$; with $p = 4$ the asymptotic test gives $r = 1$ and the wild bootstrap rejects nothing

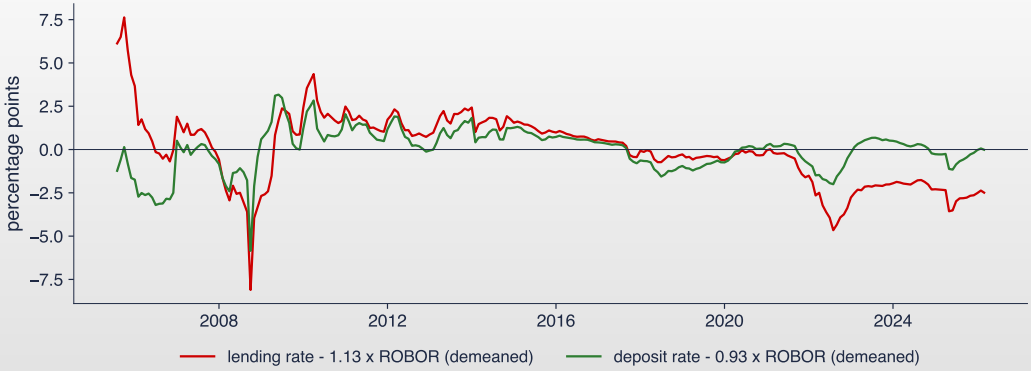
Interpreting the rank tests

- Two relations are what theory predicts: one for lending rates, one for deposit rates, both anchored to ROBOR
- The result is fragile: two extra lags and heteroskedasticity-robust inference remove the evidence; the 2008–2009 volatility dominates the sample
- We continue with $r = 2$ because theory and the parsimonious model agree, and we report the sensitivity
- A rank chosen this way is a modelling decision, not a measurement: state it before testing restrictions

Identified relations and the tests

- Just-identified (one exclusion per vector): $\text{lend}_t = 1.13\,\text{ROBOR}_t + 3.62$, $\text{dep}_t = 0.93\,\text{ROBOR}_t + 0.09$
- Adjustment: lending rate $\alpha_{11} = -0.056$ (to its own relation), deposit rate $\alpha_{22} = -0.071$; ROBOR: 0.014 and 0.017
Hypothesis	LR	df	p
complete pass-through to both rates	8.98	2	0.011
complete pass-through to the lending rate	4.16	1	0.041
ROBOR weakly exogenous (zero row of α)	0.57	2	0.752
- With $p = 4$: lending pass-through 1.11, p-value of complete pass-through to lending 0.458, of weak exogeneity 0.819

The two equilibrium errors



- $\hat{\beta}_i' y_t$ for the just-identified relations, demeaned; they should look stationary if the rank and β are right

ATS_ch4_passthrough_vecm

Interpreting the pass-through VECM

- Lending rates over-react slightly in the long run (1.13 per point of ROBOR), deposit rates under-react (0.93): the bank margin widens when ROBOR rises
- Joint complete pass-through is rejected ($p = 0.011$), driven by deposits; for lending rates alone the verdict depends on the lag length
- ROBOR is weakly exogenous ($p = 0.752$): money-market rates lead, bank rates follow; this justifies a single-equation ARDL
- The lending equilibrium error is persistent before 2010: the transition-era spread is not a stable markup, a candidate for a broken constant Johansen, Mosconi and Nielsen (2000)

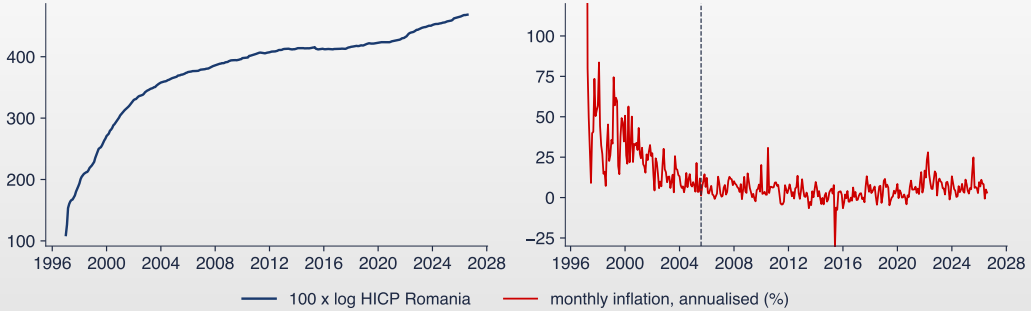
Recap: Pass-through in Romania

- Two cointegrating relations with ROBOR as the common trend, but the rank is sensitive to lags and to heteroskedasticity
- Lending pass-through close to (slightly above) one, deposit pass-through below one
- Weak exogeneity of ROBOR is not rejected: a conditional single-equation model is legitimate

Limits of the I(1) model: I(2) trends

- If $\alpha'_\perp \Gamma \beta_\perp$ has reduced rank $s < n - r$, the Granger representation fails
  - some common trends are I(2): y_t contains double sums $\sum_{j \le t} \sum_{i \le j} \varepsilon_i$
  - typical candidates: nominal prices and nominal money when inflation changes; inflation is then I(1)
- The I(2) model Johansen (1995c), Juselius (2006)
  - two reduced-rank conditions: $\Pi = \alpha\beta'$ and $\alpha'_\perp \Gamma \beta_\perp = \xi\eta'$
  - ξ, η ($(n - r) \times s$, full rank): the factors of the second reduced-rank matrix
  - polynomial cointegration: $\beta' y_t + \delta' \Delta y_t \sim$ I(0); δ ($n \times r$): the weights of the differences
  - example: real money plus a multiple of inflation
- Practical check
  - test Δy_t for a unit root; a root near one in the stationary part of the I(1) VECM is a warning
  - the usual fix: model real (deflated) variables and inflation

An I(2) check for the Romanian price level



- Left: 100 ln HICP; right: monthly inflation, annualised; dashed line: inflation targeting from August 2005
- ADF t on monthly inflation (12 lags): -4.25 for 1997–2026, -2.39 since 2005 (5% critical value -2.87); on its change: -6.13

ATS_ch4_i2_check

Interpreting the I(2) check

- Over 1997–2026, disinflation looks like mean reversion and inflation tests as stationary: the price level is I(1)
- Since 2005 a unit root in inflation is not rejected: within the inflation-targeting sample the price level behaves like I(2)
- The answer depends on the sample, as for the rank: this is why systems with nominal levels are usually rewritten in real terms and inflation
- Our pass-through system uses rates, which are at most I(1): the I(2) issue does not arise there

Permanent and transitory shocks (1/2)

- Granger representation: the long-run effect of ε_t on y_{t+h}, $h \to \infty$, is $C\varepsilon_t$, and $\operatorname{rank} C = n - r$
- Structural shocks: $\varepsilon_t = Bu_t$, $\mathbb{E}u_t u_t' = I$
  - u_t: n uncorrelated shocks with unit variance; B ($n \times n$): their impact effects, $BB' = \Omega$
  - the long-run impact CB has at most $n - r$ non-zero columns: $k = n - r$ permanent and r transitory shocks
- Counting (Chapter 3 needed $n(n-1)/2$ restrictions): cointegration gives rk zero long-run restrictions for free
  - still needed: $k(k-1)/2$ among the permanent shocks and $r(r-1)/2$ among the transitory shocks Kilian and Lütkepohl (2017), Gonzalo and Ng (2001)

Permanent and transitory shocks (2/2)

- With one common trend ($k = 1$) the permanent shock is identified without further assumptions
  - the shock: $u_t^P = \alpha_\perp' \varepsilon_t / (\alpha_\perp' \Omega \alpha_\perp)^{1/2}$, the innovation of the common trend scaled to unit variance
  - its impact: $b^P = \Omega \alpha_\perp (\alpha_\perp' \Omega \alpha_\perp)^{-1/2}$, the column of B that moves the n variables on impact
- This generalises Blanchard–Quah (Chapter 3): there the long-run zero was imposed; here it comes from the cointegration rank

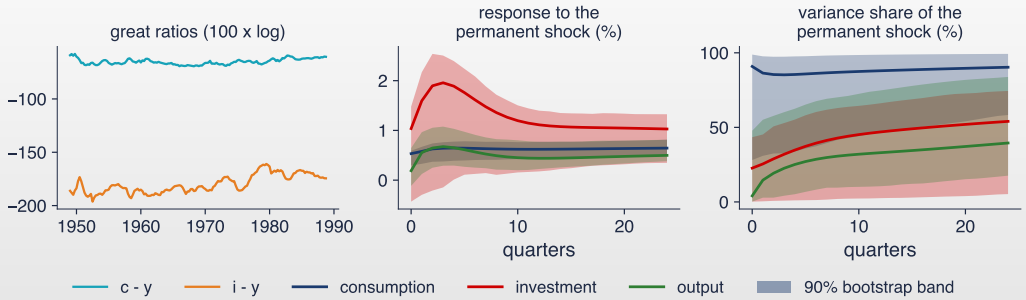
Case study: King, Plosser, Stock and Watson (1991) (1/2)

- King, Plosser, Stock and Watson (1991)
  - a real business-cycle model with a stochastic productivity trend
  - consumption, investment and output share one common trend
  - the great ratios $c - y$ and $i - y$ are stationary (c, i, y: logarithms per capita)
- In VECM terms
  - $n = 3$, $r = 2$, $\beta = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{pmatrix}$ imposed for (c, i, y)
  - $k = n - r = 1$: one permanent "balanced-growth" shock
- Question: how much of the business cycle does the permanent shock explain?

Case study: King, Plosser, Stock and Watson (1991) (2/2)

- Our replication: US quarterly data, 1949Q1–1988Q4 (the paper's sample), $T = 160$
  - c: nondurables and services; i: fixed investment; y: GDP
  - all per capita, deflated by the GDP deflator
- Rank
  - VAR(2) in levels (all criteria), case 3
  - trace: 45.8, 19.4, 1.4 against 95% quantiles 29.8, 15.5, 3.9: $r = 2$
- Test of $\beta = \beta_0$ (the exact great ratios)
  - case 3: rejected, $LR = 16.6$, $\chi^2(2)$, $p < 0.001$
  - with a restricted trend (case 4): not rejected, $LR = 0.50$, $p = 0.778$

The balanced-growth shock



- Great ratios imposed (case 3); one-standard-deviation permanent shock; 90% residual-bootstrap bands (300 samples, β fixed)
- Long-run effect 0.74% on all three; share in output variance: 4% at 1 quarter, 30% at 8, 39% at 24 [17; 83]

ATS_ch4_common_trends

Interpreting the common-trends model

- Balanced growth holds by construction in the long run: the shock moves c, i and y by the same 0.74%; investment overshoots in the short run
- The permanent shock dominates consumption (87% at 8 quarters, the permanent-income logic) but explains a minority of output fluctuations at business-cycle horizons
- Bands are wide: with $T = 160$ the data barely separate permanent from transitory variation
- As in Chapter 3, the economics is in the identifying assumption: here, which relations are stationary

Robustness of the common-trends results

Specification	$h = 1$	$h = 4$	$h = 8$	$h = 24$
great ratios, case 3, 1949–1988	4%	23%	30%	39%
great ratios with trend, case 4, 1949–1988	34%	37%	52%	77%
great ratios, case 3, 1949–2019	51%	30%	29%	40%
great ratios, case 3, 1949–2025	28%	20%	20%	29%

- Share of the permanent shock in the forecast error variance of output, h quarters ahead
- Allowing the ratios to trend (services gaining share) raises the role of the permanent shock considerably at short horizons; extending the sample changes it again; the great-ratio LR rises to 24.3 on data to 2025
- Conclusion: the size of "trend shocks" in the cycle is not pinned down by the data alone

Recap: Structural VECM

- Cointegration splits shocks into $n - r$ permanent and r transitory ones and supplies rk long-run zeros
- With one common trend the permanent shock is $\alpha'_\perp \varepsilon_t$, scaled
- KPSW: balanced growth gives a clean identification, but the variance shares depend on the deterministic case and the sample

From a VECM to a conditional ECM (1/2)

- Partition $y_t = (y_t, x_t')'$, x_t of dimension k; if x_t is weakly exogenous and there is one relation involving y_t, the conditional model is

$$\Delta y_t = c_0 + c_1 t + \pi_{yy} y_{t-1} + \pi_{yx}' x_{t-1} + \sum_{i=1}^{p-1} \psi_i \Delta y_{t-i} + \sum_{j=0}^{q-1} \omega_j' \Delta x_{t-j} + u_t$$

- Notation Pesaran, Shin and Smith (2001)
  - y_t: the dependent variable; x_t: the k forcing variables; c_0, c_1: intercept and trend coefficient
  - π_{yy} (scalar), π_{yx} $(k \times 1)$: the coefficients on the lagged levels, i.e. the long-run part
  - ψ_i, ω_j: short-run coefficients; p, q: lag orders; u_t: the error, uncorrelated with Δx_t

From a VECM to a conditional ECM (2/2)

- This is an ARDL(p, q) in levels, rewritten
  - long-run coefficients $\theta = -\pi_{yx}/\pi_{yy}$: the equilibrium is $y = \theta' x$ (plus constant)
  - speed of adjustment $\pi_{yy} \in (-1, 0)$: the share of last period's disequilibrium corrected this period; half-life $\ln 0.5 / \ln(1 + \pi_{yy})$
- Inference
  - OLS is consistent and $\hat{\theta}$ has a mixed-Gaussian limit when the lags absorb the endogeneity of x_t Pesaran and Shin (1998)
  - standard errors of $\hat{\theta}$ by the delta method; t-tests on θ are asymptotically valid
- The attraction: x_t may be I(0), I(1) or a mix, and we need not pre-test it

The bounds test (1/2)

- H_0: no level relationship, $\pi_{yy} = 0$ and $\pi_{yx} = 0$
  - F: the Wald statistic of these $k + 1$ restrictions, divided by $k + 1$
  - t: the t ratio of $\hat{\pi}_{yy}$; it excludes the degenerate case $\pi_{yy} = 0 \neq \pi_{yx}$
- The null distribution depends on the integration order of x_t
  - lower bound: all of x_t is I(0); upper bound: all of x_t is I(1)
- Decision
  - F above the upper bound: reject, there is a level relationship
  - F below the lower bound: do not reject
  - between the bounds: inconclusive, the order of integration of x_t matters

The bounds test (2/2)

- Five cases, as in Johansen
  - I: no deterministic terms; II: restricted intercept
  - III: unrestricted intercept; IV: unrestricted intercept, restricted trend
  - V: unrestricted intercept and trend
  - in cases II and IV the restricted terms are part of H_0; the t-bounds exist for cases I, III, V
- Simulated 5% bounds, $k = 1$ ($T = 1000$, 10 000 replications)
  - F: case II [3.58; 4.14], case III [4.97; 5.71]
  - t, case III: $[-2.86; -3.22]$

Critical values in small samples

- PSS tabulate asymptotic bounds ($T = 1000$); with 30–80 annual observations they are too low Narayan (2005)
- Our simulation, case III, $k = 1$, 5%: $T = 30$: [5.52; 6.32], $T = 80$: [5.02; 5.86], $T = 250$: [4.98; 5.73], asymptotic: [4.97; 5.71]
- Response-surface regressions give critical values and approximate p-values for any T, k and lag order Kripfganz and Schneider (2020): use them instead of the asymptotic table
- Frequent misuses in applied work
  - the wrong case (case III bounds for a model estimated in case II); I(2) regressors, for which the bounds are invalid
  - feedback from y_t to x_t (no weak exogeneity), or more than one relation among the variables

Choosing between ARDL and VECM

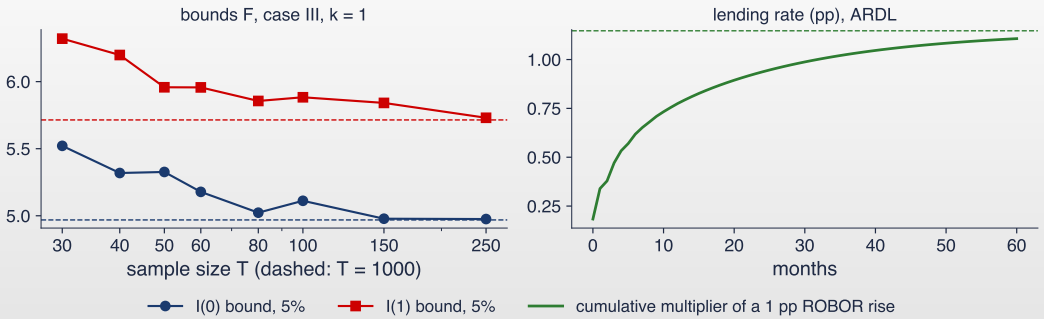
	ARDL	VECM
Relations	one, normalised on y_t	r, rank estimated
Regressors	I(0), I(1) or mixed; weakly exogenous	all I(1); all endogenous
Inference	bounds F and t; θ by the delta method	trace/max-eigenvalue; χ^2 tests on β, α
Small samples	few parameters; small-sample bounds needed	many parameters; bootstrap needed
Structural use	multipliers of x on y	common trends, identified shocks

- Rule: estimate the VECM first when you can
  - weakly exogenous regressors and a unique relation: the ARDL is efficient and simpler
  - feedback from y_t to x_t: the ARDL estimates are inconsistent

ARDL pass-through of ROBOR to the lending rate

- ARDL(4, 2) selected by BIC, case II (ROBOR weakly exogenous in the VECM), $n = 244$
- Bounds $F = 4.27$; 5% bounds asymptotic [3.58; 4.14], at our T [3.67; 4.14]: a level relationship
  - the same regression in case III: $F = 5.87$ against [4.94; 5.79]: inconclusive; $t = -3.28$ against [−2.86; −3.22]
- Long-run pass-through $\hat{\theta} = 1.15$ (SE 0.12): complete pass-through not rejected by the t-test; speed $\hat{\pi}_{yy} = -0.026$ (SE 0.008), half-life 27 months
- Deposit rate: $F = 7.68$, $\hat{\theta} = 0.94$ (SE 0.05), half-life 12 months: incomplete but faster

Small-sample bounds and the dynamic multipliers



- Left: simulated 5% bounds, case III, $k = 1$, against T (dashed: $T = 1000$); right: cumulative response of the lending rate to a permanent 1 pp rise of ROBOR (dashed: $\hat{\theta}$)
- Multipliers: 0.18 on impact, 0.62 after 6 months, 0.77 after 12, 0.94 after 24, 1.11 after 60

ATS_ch4_ardl_bounds

Interpreting the ARDL results

- The upper bound falls from 6.32 at $T = 30$ to 5.73 at $T = 250$: with annual data the asymptotic table over-rejects
- The verdict on a level relationship depends on the case: decide case II or III from economics (no drift in rates) before looking at F
- ARDL and VECM agree on the size of the lending pass-through (1.15 and 1.13) but not on the test of $\theta = 1$: different nuisance parameters, different power
- About half of the long-run effect arrives within six months (0.62); the rest is slow, with a half-life of the disequilibrium of 27 months, consistent with fixed-rate periods and contract repricing

Toda–Yamamoto causality and nonlinear ARDL (1/2)

- **Toda–Yamamoto causality** Toda and Yamamoto (1995)
  - a VAR of possibly integrated or cointegrated variables
  - fit $p + d_{\max}$ lags in levels and test only the first p; $d_{\max}$: the highest suspected order of integration
  - the Wald statistic is asymptotically $\chi^2(p)$ whatever the integration and cointegration properties
  - the price: lower power than a correctly specified VECM
- **Nonlinear ARDL** Shin, Yu and Greenwood-Nimmo (2014)
  - partial sums of increases: $x_t^+ = \sum_{j \le t} \max(\Delta x_j, 0)$
  - partial sums of decreases: $x_t^- = \sum_{j \le t} \min(\Delta x_j, 0)$, so that $x_t = x_0 + x_t^+ + x_t^-$

Toda–Yamamoto causality and nonlinear ARDL (2/2)

- Nonlinear ARDL: estimation and tests
  - x_t^+ and x_t^- enter the ARDL as two regressors, with long-run coefficients θ^+, θ^- and short-run coefficients ω_j^+, ω_j^-
  - long-run asymmetry: $\theta^+ \neq \theta^-$; short-run asymmetry: different ω_j^+ and ω_j^-
  - both tested by Wald tests; the bounds test uses $k = 2$
- Classic use: "rockets and feathers" Bacon (1991)
  - retail fuel prices rise fast with oil and fall slowly
  - Seminar 4 tests it for Romania

Recap: ARDL and the bounds test

- The conditional ECM is an ARDL rewritten; it needs weak exogeneity and a single relation
- The bounds test avoids pre-testing the regressors, at the price of an inconclusive zone
- Use the right case and small-sample (response-surface) critical values
- Toda–Yamamoto for causality without pre-testing; NARDL for asymmetric adjustment

Panel data: advantages and new problems



The ECB seat, Frankfurt

Photo: DXR (2015); CC BY-SA 4.0; Wikimedia Commons

- The gain: N countries add information
  - country series are short: 20–30 years of annual data
  - a single-country rank or bounds test has little power
- Asymptotics in two dimensions Phillips and Moon (1999), Kao (1999)
  - sequential ($T \to \infty$, then $N \to \infty$) or joint, with $N/T \to 0$
  - a panel regression of I(1) series estimates an average long-run relation even without cointegration
- Two new problems
  - **heterogeneity**: the dynamics differ across countries
  - **cross-section dependence**: common shocks (2009, COVID-19, the euro)
  - first-generation tests assume independent units; second-generation tests model the dependence Breitung and Pesaran (2008)

Measuring cross-section dependence

- **CD** statistic Pesaran (2021): the scaled sum of all pairwise correlations

$$CD = \sqrt{\frac{2T}{N(N-1)}} \sum_{i<j} \hat{\rho}_{ij} \to N(0,1)$$

  - $\hat{\rho}_{ij}$: the correlation over time between the residuals (or demeaned series) of units i and j; N: number of units; T: number of periods
  - reading: $CD \approx 0$ under the null; $|CD| > 1.96$ rejects it at 5%
- The null is *weak* dependence Pesaran (2015a)
  - correlation that vanishes on average as N grows
  - a common factor with non-zero mean loadings gives $|CD| \to \infty$
  - valid for fixed T and large N, also in dynamic and unit-root panels
  - positive and negative correlations can offset each other: report the mean $|\hat{\rho}_{ij}|$ as well

Cross-section dependence in EU-27 data

- EU-27, $T = 22$ (2001–2022)
  - consumption growth: CD $= 50.3$, mean $|\hat{\rho}_{ij}| = 0.59$
  - income growth: CD $= 25.7$; inflation: CD $= 62.4$
- Interpretation
  - every statistic is far above 1.96: strong dependence everywhere
  - first-generation panel tests are not reliable for these data

Panel unit-root tests (1/2)

- **LLC** Levin, Lin and Chu (2002): one ADF regression per unit with a common ρ

$$\Delta y_{it} = \mu_i + \rho y_{i,t-1} + \sum_j \gamma_{ij} \Delta y_{i,t-j} + e_{it}$$

  - y_{it}: unit i at time t; μ_i: unit intercepts; γ_{ij}: unit-specific lag coefficients; e_{it}: errors
  - H_0: $\rho = 0$ (unit root in every unit); H_1: $\rho < 0$, all units stationary with the same ρ
  - pooled t on orthogonalised, variance-normalised residuals, adjusted by tabulated mean and variance
- **IPS** Im, Pesaran and Shin (2003): heterogeneous ρ_i; average the individual ADF statistics t_i and standardise

$$\bar{t} = N^{-1} \sum_i t_i, \qquad Z = \sqrt{N}(\bar{t} - \mathbb{E}t)/\sqrt{\mathrm{Var}(t)} \to N(0,1)$$

  - $\mathbb{E}t$, $\mathrm{Var}(t)$: the mean and variance of one ADF t under the null (tabulated); a very negative Z rejects
  - H_1: a non-zero fraction of units is stationary; rejection does not say which

Panel unit-root tests (2/2)

- **CIPS** Pesaran (2007): one common factor f_t in the errors; augment each ADF with $\bar{y}_{t-1}$ and $\Delta\bar{y}_{t-j}$ (CADF), average the t-ratios
  - $\bar{y}_t = N^{-1}\sum_i y_{it}$: the cross-section average at time t; f_t: an unobserved shock common to all units
  - the cross-section averages proxy f_t; the distribution is non-standard and tabulated
- Under strong dependence LLC and IPS over-reject: they treat 27 correlated countries as 27 independent pieces of evidence

Panel unit roots in EU-27 data

Variable	LLC	IPS $\bar{t}$	CIPS
consumption (trend)	-12.85 (0.075)	-2.67 (0.003)	-2.38 (0.290)
income (trend)	-11.82 (0.305)	-2.09 (0.660)	-2.18 (0.566)
inflation (constant)	-10.10 (0.003)	-1.82 (0.038)	-2.36 (0.009)

- c, y: 100 ln real per capita household consumption and disposable income; π: consumption-deflator inflation; $N = 27$, $T = 22$, one lag
- p-values in brackets from 1 000 simulations of the null with independent units, for our N and T (replacing the tabulated moments)
- Consumption: IPS rejects the unit root ($p = 0.003$), CIPS does not ($p = 0.290$): the "stationarity" was a common factor; inflation is stationary by every test

Panel cointegration tests (1/2)

- **Residual-based** Pedroni (1999), Pedroni (2004): estimate $y_{it} = a_i + \delta_i t + \beta_i' x_{it} + e_{it}$ country by country and test the residuals for a unit root
  - a_i, δ_i: country intercept and trend; β_i: country cointegrating coefficients; e_{it}: the equilibrium error, I(1) under H_0
  - seven statistics: four "panel" (within, common AR coefficient) and three "group" (between, heterogeneous); H_0: no cointegration in any unit
  - they impose a common-factor restriction (short-run dynamics of x and y alike); Kao (1999) is the homogeneous special case
- **ECM-based** Westerlund (2007): test whether each country's ECM has error correction

$$\Delta y_{it} = d_i + \alpha_i y_{i,t-1} + \lambda_i' x_{i,t-1} + \sum_j a_{ij} \Delta y_{i,t-j} + \sum_j \gamma_{ij}' \Delta x_{i,t-j} + e_{it}$$

  - α_i: the speed of adjustment of country i; λ_i: its level coefficients on x; d_i, a_{ij}, γ_{ij}: deterministic and short-run terms
  - H_0: $\alpha_i = 0$ for all i (no error correction anywhere); negative statistics reject

Panel cointegration tests (2/2)

- Westerlund statistics
  - group: $G_t = N^{-1}\sum_i \hat{\alpha}_i/\mathrm{SE}(\hat{\alpha}_i)$, $G_a = N^{-1}\sum_i T\hat{\alpha}_i/\hat{\alpha}_i(1)$; H_1: some countries adjust
  - $\hat{\alpha}_i(1) = 1 - \sum_j \hat{a}_{ij}$: scales $\hat{\alpha}_i$ to the long-run adjustment
  - panel: P_t, P_a, the same ratios for a common α; H_1: all countries adjust
  - bootstrap critical values under cross-section dependence
- EU-27, c on y and π (null distributions simulated)
  - Pedroni group ADF (the mean of the country ADF t-ratios of $\hat{e}_{it}$): -3.05, 5% critical value -2.73, $p < 0.001$
  - Westerlund G_t: -1.66, 5% critical value -2.18, $p = 0.714$
- The two families disagree
  - the residual-based test rejects, the ECM test does not
  - with $T = 22$ and strong dependence, neither verdict is secure

Estimators of the long-run relation (1/3)

- **Panel FMOLS/DOLS** Pedroni (2001)
  - FMOLS: fully modified OLS, corrected for endogeneity and serial correlation Phillips and Hansen (1990)
  - DOLS: OLS with leads and lags of Δx Saikkonen (1991), Stock and Watson (1993)
  - estimated per country and averaged (group mean)
- **Mean group (MG)** Pesaran and Smith (1995)
  - estimate each country's ARDL and average the long-run coefficients: $\hat{\theta}_{MG} = N^{-1} \sum_i \hat{\theta}_i$
  - consistent under full heterogeneity; SE $= s_\theta / \sqrt{N}$, s_θ: the cross-country standard deviation of $\hat{\theta}_i$
  - pooled fixed effects with common slopes are inconsistent when the true slopes differ, even for large T; for small T the Nickell bias adds Nickell (1981)

Estimators of the long-run relation (2/3)

- **CCE** Pesaran (2006)
  - model: $y_{it} = \alpha_i + \beta_i' x_{it} + \gamma_i' f_t + e_{it}$
  - estimation: add the cross-section averages $\bar{y}_t, \bar{x}_t$ to each regression
  - f_t: unobserved common factors; γ_i: the country loadings on them; the averages $\bar{y}_t, \bar{x}_t$ absorb f_t
  - valid with non-stationary factors Kapetanios, Pesaran and Yamagata (2011); dynamic version CS-ARDL Chudik and Pesaran (2015)

Estimators of the long-run relation (3/3)

- **Pooled mean group (PMG)** Pesaran, Shin and Smith (1999): a common long run inside country-specific ECMs

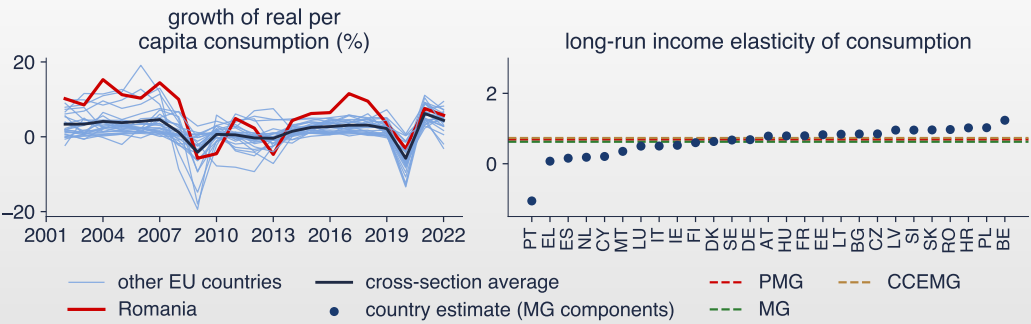
$$\Delta y_{it} = \phi_i(y_{i,t-1} - \theta' x_{i,t-1}) + \sum_j \lambda_{ij}\Delta y_{i,t-j} + \sum_j \delta'_{ij}\Delta x_{i,t-j} + \mu_i + \varepsilon_{it}$$

  - θ: the long-run coefficients, common to all countries; ϕ_i: the country speed of adjustment, $\phi_i < 0$ required
  - λ_{ij}, δ_{ij}, μ_i, $\sigma_i^2 = \mathrm{Var}\varepsilon_{it}$: short run, intercept and error variance, all country-specific
  - ML: maximise $-\sum_i \frac{T_i}{2} \ln \hat{\sigma}_i^2(\theta)$ over θ; T_i: the sample length of country i (derivation: Appendix)
- Hausman test of $\theta_{MG} = \theta_{PMG}$
  - $H = (\hat{\theta}_{MG} - \hat{\theta}_{PMG})'[\widehat{\mathrm{Var}}(\hat{\theta}_{MG}) - \widehat{\mathrm{Var}}(\hat{\theta}_{PMG})]^{-1}(\hat{\theta}_{MG} - \hat{\theta}_{PMG}) \to \chi^2(\dim\theta)$
  - PMG is efficient under homogeneity, MG consistent in both cases: a small H (large p-value) supports pooling

Case study: Pesaran, Shin and Smith (1999)

- Pesaran, Shin and Smith (1999)
  - consumption functions for OECD countries, annual data
  - ARDL in log real per capita consumption, log real per capita disposable income and inflation
  - they compare MG, PMG and dynamic fixed effects and use Hausman tests to decide whether the long-run income and inflation effects can be pooled
  - their argument: theory (permanent income) restricts the long run; adjustment speeds, habits and credit constraints differ across countries
- Our replication of the specification on EU-27, 2001–2022, ARDL(1,1,1) for every country, 672 observations
  - c_{it} on y_{it} and π_{it} as defined above; national currencies (country intercepts absorb the units)
- We start in 2000 to avoid the inflation episodes of the 1990s in Bulgaria and Romania, which would dominate a pooled inflation coefficient

Heterogeneity across the EU



- Left: consumption growth (Romania in red, cross-section average in dark blue); right: country long-run income elasticities from the individual ARDLs, with MG, PMG and CCEMG
- Country elasticities range from -1.06 to 1.23; Romania 0.97

ATS_ch4_panel

Long-run estimates for EU-27

Estimator	θ_y	θ_π	ϕ
MG	0.63 (0.09)	0.82 (0.32)	−0.43 (0.032)
PMG	0.69 (0.04)	0.41 (0.18)	−0.31 (0.028)
dynamic FE	0.81 (0.03)	0.01 (0.23)	−0.29 (0.025)
group-mean DOLS	0.77 (0.05)	−0.24 (0.17)	–
CCEMG	0.73 (0.05)	0.10 (0.10)	–

- Standard errors in brackets (MG, DOLS, CCEMG: dispersion of country estimates; PMG: likelihood; FE: clustered by country)
- Hausman MG against PMG: $H = 2.41$, $\chi^2(2)$, $p = 0.300$; CD of the PMG residuals: 30.9

Interpreting the panel estimates

- The income elasticity lies between 0.63 (MG) and 0.81 (FE), clearly below one: consumption does not track income one for one over 2001–2022
- The Hausman test does not reject pooling, so PMG is the efficient choice for the income effect
- The inflation effect is not robust: its sign and size change with the estimator; do not report one of them alone
- PMG residuals remain strongly dependent (CD = 30.9): standard errors that ignore common shocks are too small; CCEMG is the safer benchmark

Panel VAR: model and estimation

- $y_{it} = \mu_i + \sum_{j=1}^{p} A_j y_{i,t-j} + \varepsilon_{it}$: common dynamics, country fixed effects Holtz-Eakin, Newey and Rosen (1988)
  - y_{it}: the vector of variables of country i; A_j: lag matrices common to all countries; μ_i: country fixed effects; ε_{it}: innovations
- With small T the within estimator is biased by $O(1/T)$ (Nickell); estimate by GMM on forward-orthogonal deviations with lagged levels as instruments Abrigo and Love (2016)
- Identification of shocks as in Chapter 3 (Cholesky, signs); impulse responses and FEVD are common to all countries
- Homogeneity of A_j is the strong assumption: with macro panels of moderate T, mean-group VARs or CCE-augmented VARs are the heterogeneous alternatives

Recap: Panel time series

- Test for cross-section dependence first; with strong dependence use CIPS and CCE, not LLC/IPS and pooled OLS
- Panel cointegration tests differ in their null and in the common-factor restriction; disagreement is informative
- MG for heterogeneity, PMG when theory pools the long run, Hausman to decide, CCE against common factors

An open question

- Is the long-run pass-through of money-market rates to bank lending rates complete in Romania, and did it change after the 2022 tightening?
  - formal: H_0: $\theta_{\text{lend}} = 1$ in a rank-2 VECM with ROBOR weakly exogenous; against a break in θ or in the markup in 2022
  - falsified by a robust rejection: across pre-registered lag lengths, deterministic cases and bootstrap inference
- Why it matters: the speed and completeness of pass-through decide how much the BNR must move its rate to move credit conditions
  - literature to start from: de Bondt (2005), Égert, Crespo-Cuaresma and Reininger (2007), Johansen (1995a), Cavaliere, Rahbek and Taylor (2012), Pesaran, Shin and Smith (2001)

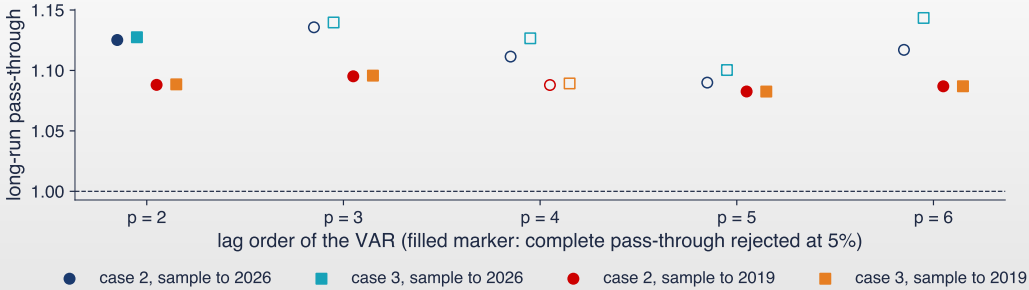
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
  - **literature**: `List peer-reviewed studies of interest-rate pass-through in Central and Eastern Europe that use cointegrated VARs or ARDL; give DOIs.` Then check every DOI on Crossref
  - **hypothesis**: `Propose three reasons why lending-rate pass-through could exceed one, and one testable implication of each.`
  - **code and replication**: ask for a Johansen function with restricted constant, then reproduce a known number first (the eigenvalues of this lecture)
  - **robustness and critique**: `Act as a hostile referee: list every specification choice that could flip the complete pass-through test.`
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

Required checks

- Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- The deterministic case of the critical values matches the estimated model (a frequent AI error: case-3 values for a case-2 model)
- The degrees of freedom of every LR test on β and α are counted by hand
- Lag length, case, sample and the treatment of 2008–2009 are fixed before the results; all variants are reported
- A claim of "complete pass-through" is checked against the confidence set of θ, not against one p-value

Mini-case: the long-run pass-through test across specifications



- Long-run pass-through to the lending rate in the rank-2 VECM: lags $p = 2, \ldots, 6$, cases 2 and 3, samples ending in 2019 and in 2026; filled markers: $\theta = 1$ rejected at 5%
- All 20 estimates lie between 1.08 and 1.14, yet complete pass-through is rejected in 10 of them: an AI summary that reports one p-value "with confidence" is wrong

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Project idea

- **Interest-rate pass-through in Central and Eastern Europe**: replicate first, then extend
  - replicate: the Romanian rank-2 VECM of this lecture (eigenvalues 0.244, 0.177, 0.017) and the ARDL pass-through 1.15
  - extend: Romania, Hungary, Czechia and Poland; broken constants Johansen, Mosconi and Nielsen (2000) in 2008 and 2022; wild-bootstrap rank tests; PMG across countries with a Hausman test; NARDL for asymmetric pass-through
  - pre-register: variables, sample, case, lag rule, rank procedure, the restrictions to test and the multiple-testing correction
- Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- Johansen = reduced-rank regression; the rank test is non-standard, depends on the case and over-rejects in small samples
- Given the rank, restrictions on β (economics) and α (weak exogeneity) are ordinary χ^2 tests
- Cointegration supplies long-run zeros for structural analysis: permanent and transitory shocks
- ARDL is a conditional VECM: valid under weak exogeneity and a single relation, with small-sample bounds
- Panels add power but bring heterogeneity and cross-section dependence: CD first, then CIPS, MG/PMG and CCE

Self-assessment

Questions

- Why does one eigen-decomposition give the ML estimator of β for every rank?
- Which deterministic case fits two interest rates without drift?
- How many degrees of freedom has the test of a known β with $n_1 = 3$, $r = 2$?
- What does weak exogeneity of ROBOR allow you to do?
- Why can IPS reject a unit root that CIPS does not reject?

Next:

- Bayesian VAR, factor models and nowcasting
- Shrinkage handles the parameter problem we met in small samples; factors generalise the cross-section averages of CCE

Appendix: from the likelihood to the eigenvalues

- For fixed β: $|\hat{\Omega}(\beta)| = |S_{00} - S_{01}\beta(\beta' S_{11}\beta)^{-1}\beta' S_{10}|$; apply the partitioned determinant to $\begin{pmatrix} S_{00} & S_{01}\beta \\ \beta' S_{10} & \beta' S_{11}\beta \end{pmatrix}$ in both orders
- $|\hat{\Omega}(\beta)|\,|\beta' S_{11}\beta| = |S_{00}|\,|\beta'(S_{11} - S_{10}S_{00}^{-1}S_{01})\beta|$, which gives the ratio on the slide
- Write $\beta = S_{11}^{-1/2}\eta$: the ratio becomes $|\eta'(I - M)\eta|/|\eta'\eta|$ with $M = S_{11}^{-1/2}S_{10}S_{00}^{-1}S_{01}S_{11}^{-1/2}$; it is minimised by the eigenvectors of M with the r largest eigenvalues
- The minimum is $\prod_{i \le r}(1 - \lambda_i)$; nested ranks share eigenvectors, so the LR statistics are sums of $-T\ln(1 - \hat{\lambda}_i)$

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Appendix: deriving the long-run impact matrix C

- Multiply the VECM by $\alpha'_\perp$: $\alpha'_\perp \Delta y_t = \alpha'_\perp \sum_i \Gamma_i \Delta y_{t-i} + \alpha'_\perp \varepsilon_t$; summing gives $\alpha'_\perp \Gamma y_t \approx \alpha'_\perp \sum_{s \leq t} \varepsilon_s$ up to stationary terms
- Decompose $y_t = \beta(\beta'\beta)^{-1}\beta' y_t + \beta_\perp(\beta'_\perp\beta_\perp)^{-1}\beta'_\perp y_t$; the first part is stationary
- Hence $\alpha'_\perp \Gamma \beta_\perp \, (\beta'_\perp \beta_\perp)^{-1}\beta'_\perp y_t \approx \alpha'_\perp \sum_s \varepsilon_s$; invert $\alpha'_\perp \Gamma \beta_\perp$ (the I(1) condition) and premultiply by $\beta_\perp$
- Result: the non-stationary part of y_t is $\beta_\perp(\alpha'_\perp \Gamma \beta_\perp)^{-1}\alpha'_\perp \sum_s \varepsilon_s = C \sum_s \varepsilon_s$

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Appendix: the PMG likelihood

- $\ell(\theta, \phi, \sigma^2) = -\sum_{i=1}^{N} \frac{T_i}{2} \ln(2\pi\sigma_i^2) - \sum_{i=1}^{N} \frac{1}{2\sigma_i^2} (\Delta y_i - \phi_i \xi_i(\theta))' H_i (\Delta y_i - \phi_i \xi_i(\theta))$
- Notation
  - Δy_i, $y_{i,-1}$ ($T_i \times 1$), $X_{i,-1}$ ($T_i \times k$): the changes, the lagged levels of y and of x for country i
  - $\xi_i(\theta) = y_{i,-1} - X_{i,-1}\theta$: the equilibrium error of country i
  - $H_i = I - W_i(W_i'W_i)^{-1}W_i'$; W_i: the country's short-run regressors and intercept, which H_i projects out
- Given θ, ϕ_i and σ_i^2 are country OLS; the concentrated likelihood $-\sum_i \frac{T_i}{2} \ln \hat{\sigma}_i^2(\theta)$ is maximised over θ (back-substitution or Newton)
- $\hat{\theta}_{PMG}$ is consistent and asymptotically normal if $\phi_i < 0$ for all i and the long run is homogeneous; the I(0)/I(1) status of x does not matter, as in ARDL

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